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System Overview (Stage 1)

System Diagram

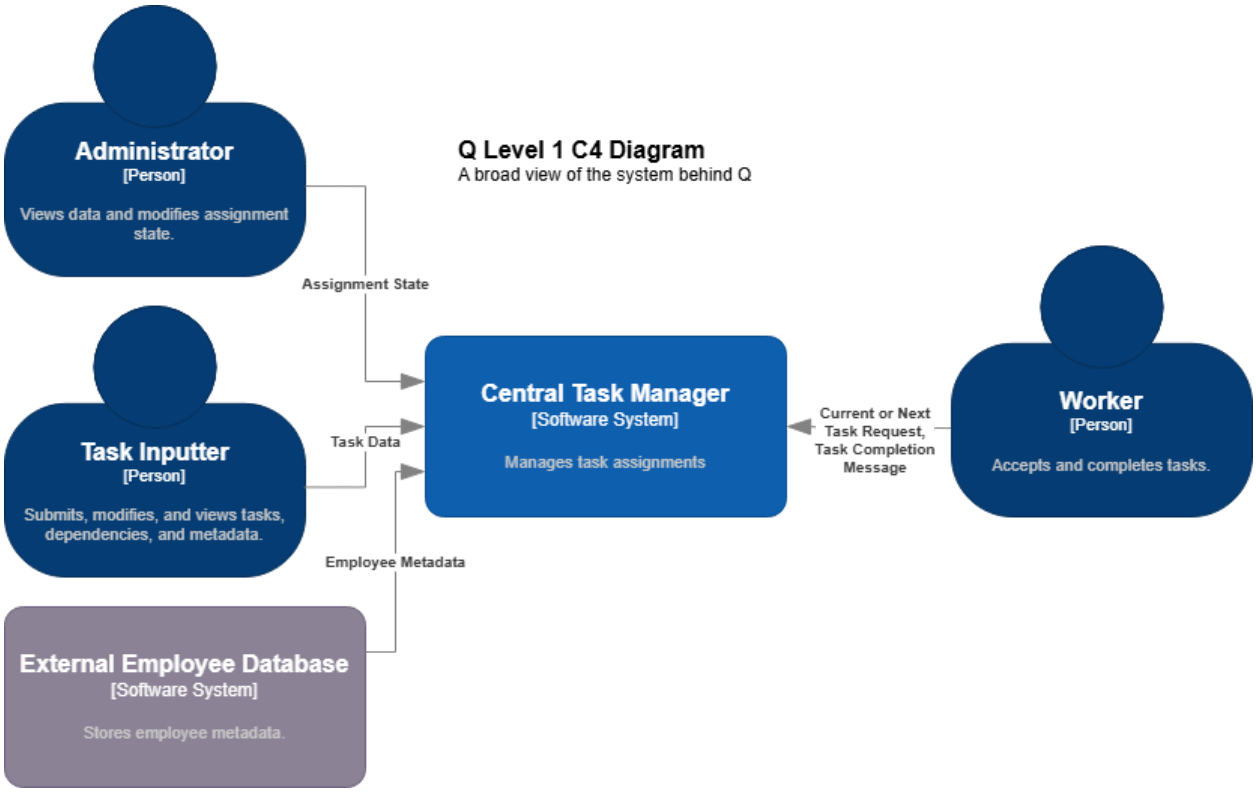


Figure 1

As demonstrated in this diagram, the central task management system interacts with the following parties:

1. Data sources (Task Inputters and External Employee Database), who provide the system with the information necessary to make its calculations.
2. Workers, the consumers of the system's calculations.
3. Administrators, who can override the system's calculations and data.

Workflow (Happy Path)

- The key objects the system keeps track of are tasks and workers.
- Therefore, the first thing a new user group must do is input tasks and workers, and any other data pertaining to them, into the system.

- When a worker is ready for work, they query the system to initiate their next task assignment and receive a task calculated by the system to be the most optimal task for them. If they have not completed their previous assignment, they will receive their current task as the response.
- A worker must mark their task as completed before they can be assigned a new task.

Failure Scenarios

- Since tasks, workers, and their metadata can be added and updated anytime, the system can't guarantee that every view of the current system data will be exact or that every task assignment will be the most optimal at the exact time of the query, but it will aim to ensure that every task assignment will be as optimal as possible. It will sacrifice speed of response over correctness because it is assumed that workers would rather get their next task faster (higher availability) at the expense of maximum optimality (stronger consistency).
- If a worker fails to receive the response containing their next task, they can simply requery the system because the calculation is stored internally.
- If, after a completion, a worker queries their next task and receives their previous task, it means their completion message was not delivered or processed successfully and they must resubmit that message.
- To ensure that a worker does not accidentally mark their next task as completed, they must submit the task ID when marking the task as completed.
- A task can be marked as completed multiple times with no adverse effects.
- Tasks may be reassigned by an administrator, e.g. to prevent lockout of a task by an unresponsive worker. The task may be either manually reassigned to a worker who has not been assigned a task yet, e.g. if they completed their previous task but haven't accepted a new one, or simply returned to the system. While this is in progress, the former assignee is locked out of the system to prevent inconsistencies. Affected workers will receive notifications if this happens. If they complete their previously assigned task during or after this process, the system will reject their completion message.